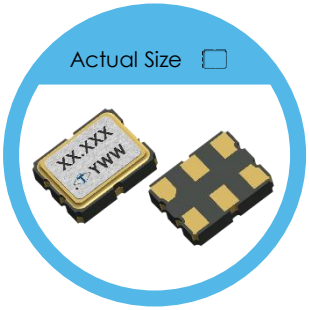
TAITIEN
Our Performance, Your Reputation

OA-M Type High Frequency up to 1.5GHz
FASTXO 3.2 x 2.5 mm SMD Differential Output Crystal Oscillator



RoHS Compliant

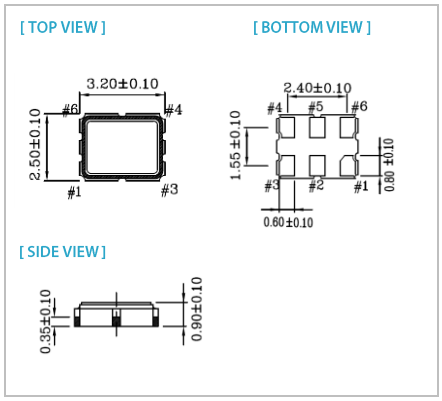
FEATURES

- Low Power Supply Voltage: 3.3V, 2.5V Supply Options
- Differential Output : LVPECL, LVDS
- Frequency Support from 10MHz to 1.5GHz
- Low Noise Typical: 0.6ps at 12kHz to 20MHz Frequency Offsets
- Temperature Range: -40 to 85°C Operation
- Pb-free/RoHS Compliant

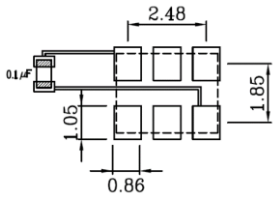
TYPICAL APPLICATION

- Enterprise Server, SAS/SATA, Microprocessors/DSP/FPGA, Broadband Access,Smart Grid
- High-Speed Gigabite Ethernet, Fiber Channel, Storages Area Network, SONET

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between V_{DD} and GND pads.

PIN FUNCTION (mm)

PIN#	FUNCTION
1	NC/Tri-State
2	Tri-State/NC
3	GND
4	Output
5	Comp. Output
6	Supply Voltage

ELECTRICAL SPECIFICATION

Parameter		LVPECL				LVDS				Unit	Test Condition
		3.3V		2.5V		3.3V		2.5V			
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD})		V_{DD} – 10%	V_{DD} + 10%	V_{DD} – 5%	V_{DD} + 5%	V_{DD} – 10%	V_{DD} + 10%	V_{DD} – 5%	V_{DD} + 5%	V	
Frequency Range		10	1500	10	1500	10	1500	10	1500	MHz	
Supply Current			50		45		45		35	mA	
Output Level	Output High	2.27	2.7	1.47	1.9		1.6		1.6	V	
	Output Low	1.45	1.7	0.65	0.9	0.9		0.9		V	
Transition Time (10%-90%)	Rise Time / Fall Time		1.0		1.0		1.0		1.0	nSec	
Duty Cycle		45	55	45	55	45	55	45	55	%	
Startup Time			10		10		10		10	mSec	
Tri-State Mode(Input to Pin 2)	Enable	0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		V	
	Disable		0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}	V	
Stand by Current			18		18		18		18	mA	
Output Loading		50Ω into V_{DD} – 2				100Ω					
Phase Noise		Typ.	Max.	Typ.	Max.	Typ.	Max.	Typ.	Max.		
At V_{DD}=3.3V, fout=250MHz	1kHz offset	-107		-107		-107		-107		dBc/Hz	
	10kHz offset	-111		-111		-111		-111		dBc/Hz	
	100kHz offset	-114		-114		-114		-114		dBc/Hz	
	1MHz offset	-125		-125		-125		-125		dBc/Hz	
	20MHz offset	-147		-147		-147		-147		dBc/Hz	
RMS Phase Jitter (12 kHz~20 MHz)(Fractional Mode)		0.8	1.5	0.8	1.5	0.8	1.5	0.8	1.5	pSec	
RMS Phase Jitter (12 kHz~20 MHz)(Integer Mode)		0.6	1.2	0.6	1.2	0.6	1.2	0.6	1.2	pSec	

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±25	±50
-10 ~ +60	○	○
-20 ~ +70	○	○
-40 ~ +85	△	○
-40 ~ +105	×	○

○: Available △:Conditional ×: Not Available
Inclusive of calibration @ 25°C ,operating temperature rrange,input Voltage variation,load variation,aging (1st year),shock,and vibration

Note: not all combination of options are available. Other specifications may be available upon request.
Specifications subject to change without notice.

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